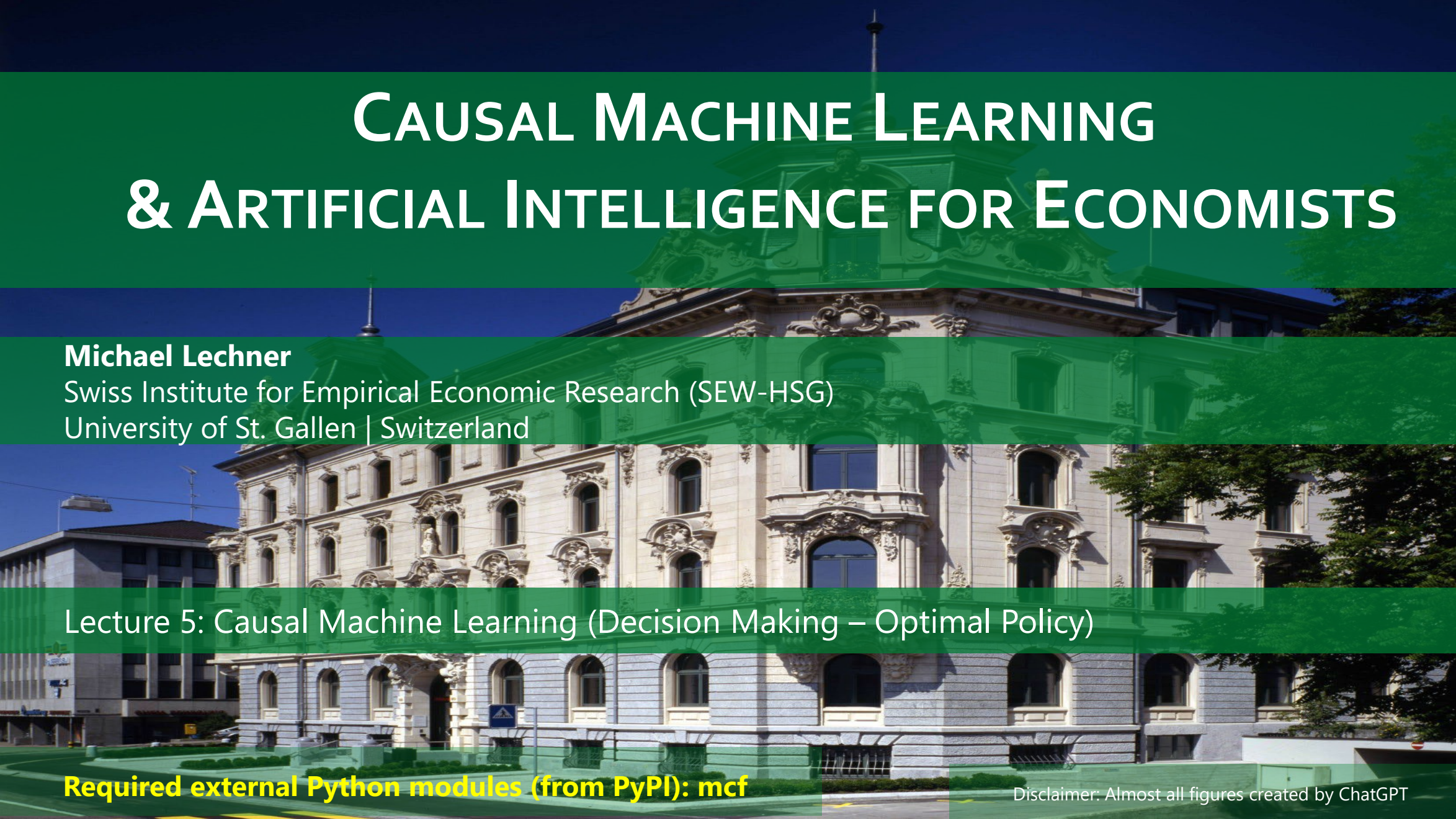
Causal Machine Learning & Artificial Intelligence for Economists

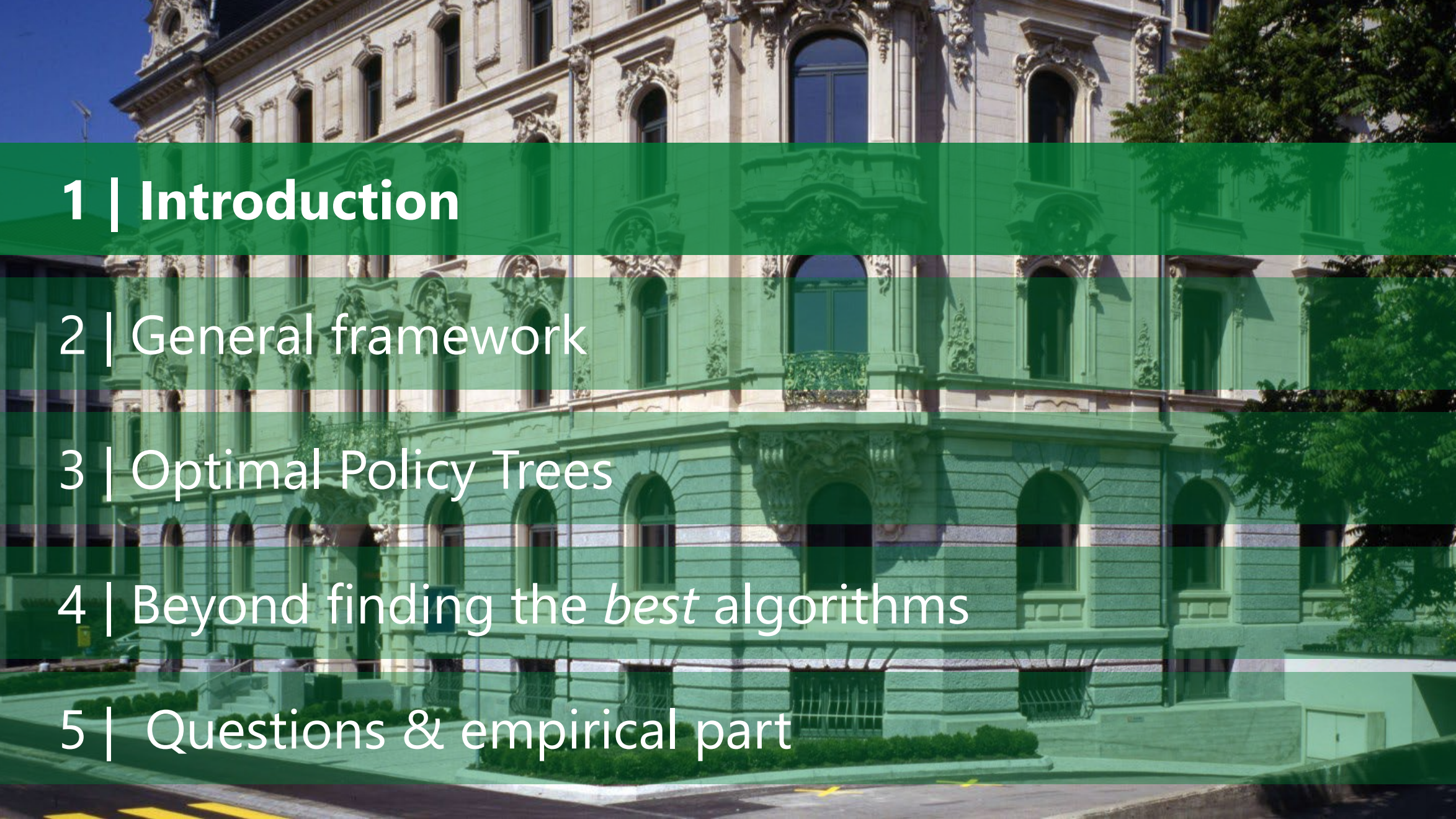


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Lecture 5: Causal Machine Learning (Decision Making – Optimal Policy)

Required external Python modules (from PyPI): mcf

Disclaimer: Almost all figures created by ChatGPT



1 | Introduction

2 | General framework

3 | Optimal Policy Trees

4 | Beyond finding the *best* algorithms

5 | Questions & empirical part



Why this topic is interesting

Many policy questions are not only about whether a programme works, they are about deciding who should receive which intervention, under constraints

Optimal policy translates causal effect estimates into allocation rules

Scarcity
budgets, places, staff time

Heterogeneity
people differ in expected gains

Accountability
rules should be explainable



From causal effects to decisions

Data
Y, D, X

—

Policy score
Potential outcomes

—

Decision rule
$\pi(X)$: who gets what?

—

Allocation
Specific treatment option

—

Outcomes
Welfare

Estimation tells us what would happen; policy learning tells us what we should do



Motivating example: Active labour-market policy

For each person, the largest expected gain may come from a different action

Caseworkers may choose among several actions: no programme, counselling, training, or subsidised employment

Person	No programme	Counselling	Training	Subsidised work
A	0	+2	**+8**	+4
B	0	+5	+1	**+9**
C	0	+1	-2	**+3**

Optimal policy: choose the action with the largest feasible gain in terms of decision maker-defined welfare (time in (un-)employment, earnings, …)



Why optimal policy is different from point estimation

Point estimation

- Target: estimate a parameter
- Example: ATE = 3.2
- Loss: estimation error, bias, variance
- Output: point estimate + standard error

Optimal policy

- Target: choose an action rule, $\pi(x)$
- Example: $\pi(x)$ chooses a programme
- Loss: welfare loss / *regret*
- Output: implementable allocation rule

different objectives

Note

***Optimal* is not a purely statistical concept. It depends on the chosen welfare criterion & the feasible set of actions**



For policy, rankings matter

A policy maker mainly needs to know which action is best for which covariate profile

Small estimation errors are harmless if they do not change the chosen action, but harmful if they reverse the ranking



Policy loss depends on choosing the wrong action, not on estimating every effect precisely

Policy learning should use a different 'quality rule' than estimation (in ML language: Policy learning is more like classification, while effect estimation is more like regression)



The general framework

A policy is a rule that maps observed characteristics into actions:

$$\pi : X \rightarrow D$$

Choose π to maximise *E[welfare under π(X)]* subject to constraints

Inputs
effects, costs, constraints

Policy class
simple rules, like trees, linear-in-X rules

Objective
value, welfare



Optimal does not mean unconstrained: The role of constraints

Policy learning is usually constrained optimization

Real-world constraints can be as important as estimated treatment effects

Budget
limited resources

Capacity
limited programme slots

Fairness
protected groups / equity

Explainability
rules humans can inspect

Legal feasibility
eligible actions only

Implementation
caseworker and institutional constraints



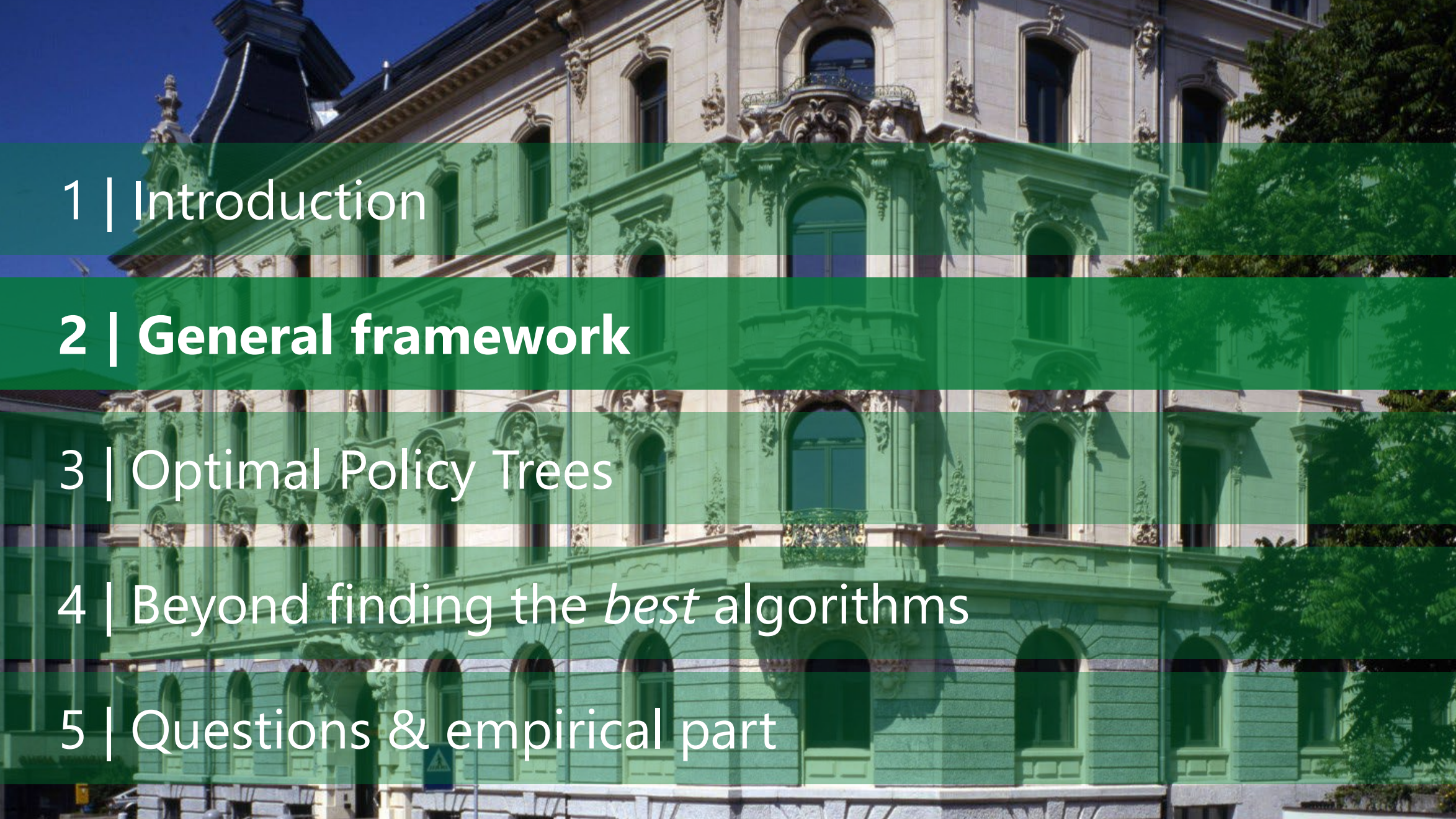
What this lecture is about

Optimal policy uses causal effect estimates to construct decision rules

The goal is not the most accurate point estimate, but a policy with high value & acceptable properties

We will focus on explainable rules, especially optimal policy trees (*human in the loop*)

1. **Introduction & motivation**
2. General framework
3. Optimal policy trees
4. Beyond finding the best algorithm


1 | Introduction
2 | General framework
3 | Optimal Policy Trees
4 | Beyond finding the *best* algorithms
5 | Questions & empirical part



Optimal policy & individualised treatment rules

Holy grail & final culmination of Causal Machine Learning

- So far: We estimated heterogeneous causal effects
- Now: We use this knowledge to improve decisions
  - This may also lead to changes in the estimation of the heterogeneous effects
  - New & rapidly developing literature in all fields

Some important papers

- *Manski (2004):* Statistical Treatment Rules for Heterogeneous Populations, *Econometrica, 72*
- *Athey, Wager (2021):* Policy Learning with Observational Data, *Econometrica*
- *Kitagawa, Tetenov (2018):* Who Should Be Treated? Empirical Welfare Maximization Methods for Treatment Choice, *Econometrica*, 86, 591–616
- *Zhou, Wager, Athey (2022):* Offline Multi-Action Policy Learning: Generalization & Optimization



Notation

$D \in \{0,1,...,M-1\}$	M exclusive treatments
$Y(m)$	potential outcome (PO) of treatment m
$Y = \sum_{m=0}^{M} \underline{1}(d=m)Y(m)$	observed outcomes
$X = (X^1,...,X^P)$	covariates (exogenous)
$p(m,x) = P(D=m \mid X=x)$	propensity scores for given value of x
$\mu(m,x) = E(Y(m) \mid X=x)$	expected PO of m for given value of x
$IATE(m,l;x) = \mu(m,x) - \mu(l,x)$	individualised average treatment effect for x
$\pi : \chi \rightarrow \{0,1,...,M-1\}$	assignment (policy) rule for given value of x
$Y(\pi(x))$	PO of policy rule $\pi(x)$ for given value of x
$Q(\pi) = E\left[Y(\pi(X))\right]$	policy value function
Π	class of feasible/allowed policy rules



From expected outcomes to the best action

For each action *m*, define the expected outcome for a person with characteristics *x*:

$$\mu(m,x) = E(Y(m) \mid X = x)$$

If there are no costs or constraints

$$\pi^{oracle}(x) = \arg\max_m \mu(m,x)$$

With action-specific costs $c_m(x)$

$$\pi^{oracle}(x) = \arg\max_m (\mu(m,x) - c(m,x))$$

Example: profile x

No programme	0.00
Counselling	0.05
Training	0.11
Subsidised work	0.08

Best feasible action: Training



How good is a policy rule?

A policy maps characteristics into actions

$$\pi : \chi \to \{0, \ldots, M-1\}$$

Value of policy π

$$Q(\pi) = E(Y(\pi(X)))$$

Equivalent decomposition

$$Q(\pi) = E\sum_m 1(\pi(X) = m)\mu(m, X)$$

Key idea

For each person, the policy selects one action.

The selected action contributes its expected potential outcome to welfare.

The policy value averages these selected outcomes over the population.

With costs, replace $\mu(m, X)$ by $\mu(m,X) - c(m,X)$.

Policy value measures the expected welfare generated by following the rule π



From policy value to empirical policy learning

Step 1: estimate scores

For each observation *i* & action *m*, construct a score for the welfare contribution of *i*

Step 2: evaluate policies

$$Q(\pi) = \frac{1}{N}\sum_i \sum_m 1(\pi(X)) = m)\hat{score}(m, X)$$

Step 3: choose the best rule

$$\pi = \arg\max_{\pi \in \Pi} Q(\pi)$$

What can the score be?

Estimated potential outcomes

or cross-fitted doubly robust policy scores.

The score should capture the expected welfare contribution of assigning action *m* to observation *i*.

Why *Π* matters (for the next section)

Π is the class of feasible policy rules.

If Π is the class of *shallow decision trees*,

the solution is an *optimal policy tree*.

Constraints such as budgets or capacities can be imposed when searching over Π.

Policy learning estimates the value of admissible rules & chooses the rule with the highest estimated value



Minimax regret as quality measure for algorithms

A policy-learning method has low minimax regret if its worst-case expected welfare loss is small over a specified class of data-generating processes, *P*:

$$\pi^{MM*} = \inf_{\hat{\pi}\in\Pi} \sup_{P\in\mathrm{P}} E_P[R_P(\hat{\pi})] \qquad R_P(\pi) = Q_P(\pi_P^*) - Q_P(\pi)$$

Minimax regret is an important theoretical benchmark for policy-learning algorithms



Online vs offline policy learning

Offline policy learning (**this lecture**)

- Experimental or non-experimental data is already available
- Use this data to learn good policy rules to apply in future
- Good approach if data is large & informative & world is stationary

Online policy learning (not considered today)

- Gather new data (usually experimental)
- Allocate observations to balance …
  - good policy (exploitation)
  - improving estimates (possibly suboptimal allocation in smallish groups; exploration)
  - This trade-off needs to be optimised to get good long-term results
    - Large & growing literature (Bandits) on how to best do this



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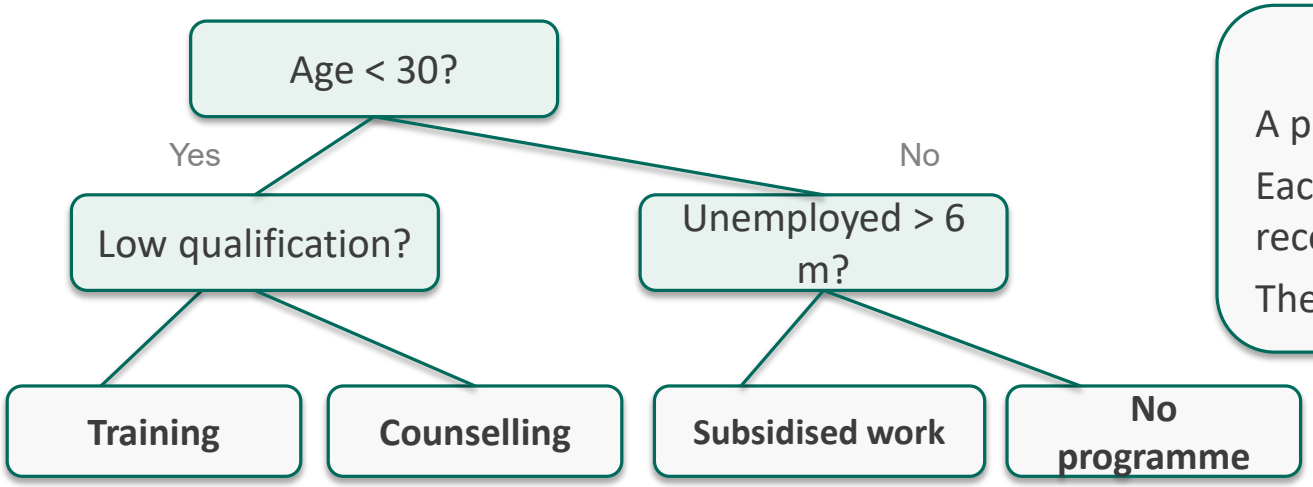


What is a policy tree?

A policy tree is a decision rule that maps observed characteristics into actions

Internal nodes ask simple yes/no questions about observed covariates

Terminal leaves assign the action to be taken for that covariate profile



Key idea

A person follows one path through the tree.

Each final leaf corresponds to a simple treatment recommendation.

The tree is an interpretable if–then decision rule.



Why policy trees are attractive

Explainability
The rule can be read as a small set of if–then statements

Implementation
Caseworkers and agencies can apply the same rule consistently

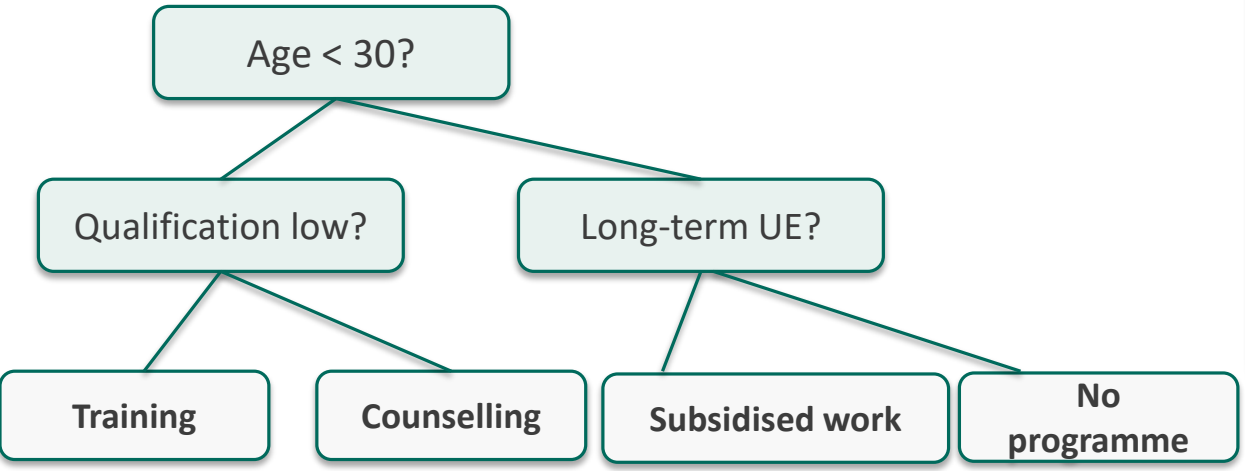
Constraints
Budgets, capacities, & eligibility rules can be incorporated

Robustness
Restricting the policy class reduces overfitting and improves credibility

Trade-off: more complex rules may fit the sample better, but simple trees are easier to validate & use



Explainability: A policy tree is a transparent sequence of rules



Equivalent if–then rule list

If age < 30 & qualification is low: assign **Training**

If age < 30 & qualification is not low: assign **Counselling**

If age ≥ 30 & long-term unemployed: assign **Subsidised work**

If age ≥ 30 & not long-term unemployed: assign **no programme**

A policy tree turns a complex allocation problem into a small, inspectable rule set



Policy tree versus causal tree

Causal tree

Splits data to find groups with different treatment effects.

Terminal leaves contain estimated effects, such as CATEs.

Output: heterogeneity description.

≠

Policy tree

Splits data to find groups for which different actions are best.

Terminal leaves contain assigned actions.

Output: implementable decision rule.

Causal trees estimate *effects*; policy trees choose *actions*



What makes a tree optimal?

Restricted policy class

Consider a class of admissible trees, for example all depth-2 trees using selected variables.

Each tree in this class is a feasible policy rule.

Estimated policy value

For every candidate tree T, compute its estimated value $\hat{Q}(T)$.

Choose the tree with the highest value within the class.

The estimated policy rule is the tree with the highest value in the training data

"Optimal" means optimal within the specific tree class, not necessarily globally optimal among all imaginable rules



Greedy split choice versus optimal full-tree search

Greedy tree construction

Choose the split with the best immediate improvement at the current node.

Fix that split and continue recursively in the daughter nodes.

This can miss a better overall tree if a modest first split allows much better later splits.

Optimal policy tree

For every candidate split, compare the values of the best complete trees that can be built underneath it.

The root split is chosen only after considering the value of the full resulting tree.

Splits and leaf actions are therefore chosen jointly, not one node at a time.

Key idea: compare complete-tree values, not just one-step gains

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How an optimal policy tree is found

Main idea: do not choose the split that looks best immediately; choose the split that leads to the best complete tree.

1 | Consider candidate first splits

At the root node, look at several possible questions.

Examples: “Age < 30?” or “Qualification low?”

→

2 | Look ahead

For each candidate split, ask:

What is the best complete tree I can build if I start here?

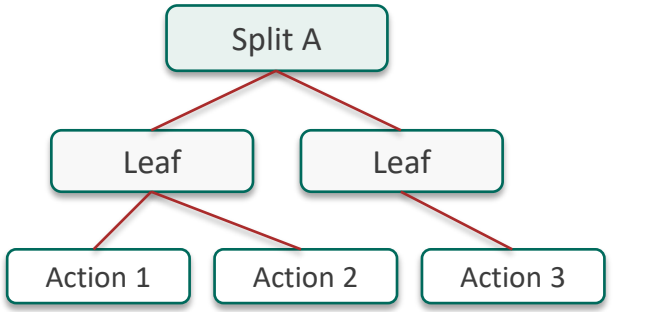
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3 | Compare complete trees

Judge each candidate by the value of the full tree below it.

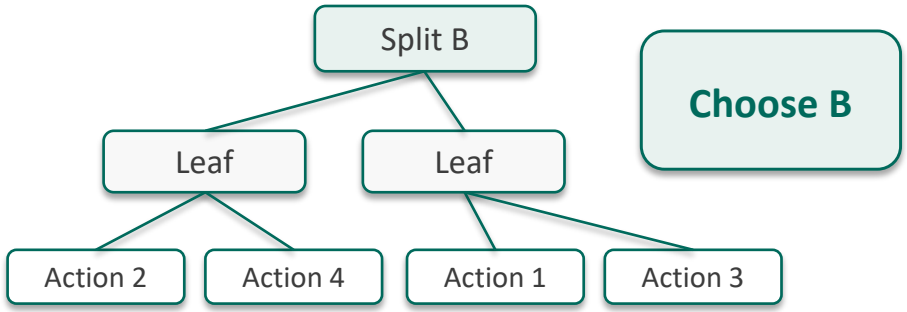
Choose the completed tree with the highest policy value.

Candidate A:
looks good immediately



Immediate gain: high | Final tree value: 8

Candidate B:
weaker first split, better final tree



Choose B

Immediate gain: smaller | Final tree value: 11

So “optimal” means: pick the root split & later splits jointly, because they are judged by the value of the full resulting policy tree



Where the technical papers enter

Step 1: causal scoring

Estimate policy scores, often based on doubly robust or cross-fitted scores.

This uses the causal-estimation machinery from the previous lectures.

Step 2: policy optimisation

Search for the best tree within a restricted class of policy trees.

For multi-action trees, Zhou–Athey–Wager provide a recursive algorithm.

Athey–Wager: binary treatment. Zhou–Athey–Wager: multiple treatments.
In both cases, the main idea is: estimate policy scores first, then optimise over a restricted policy class.



Athey & Wager (2021) | 1

Binary treatment

Selection-on-observables

Double machine learning-type-of-argument incorporated

Restricted class of policies

- Linear-in-*X*-rules (not too many *X*) & (shallow) decision trees
- Leads to much better theoretical guarantees than more complex policies
- Simple rules that depend on a limited number of covariates may be more easily accepted by decision makers ('interpretable AI')

DML & restricted class of policies leads to good theoretical guarantees



Athey & Wager (2021) | 2

AW2021 propose two-step procedure

1) Estimate doubly robust policy score

$$\hat{\Gamma}_i = \hat{\mu}(1, x_i) - \hat{\mu}(0, x_i) + \frac{d_i - \hat{p}(x_i)}{\hat{p}(x_i)\left[1 - \hat{p}(x_i)\right]}\left[y_i - \hat{\mu}(d_i, x_i)\right]$$

2) Find best allocation by minimising regret given the score

$$\hat{\pi} = \arg\max_{\pi \in \Pi} \left\{ \frac{1}{N} \sum_{i=1}^{N} \left[2\pi(x_i) - 1\right] \hat{\Gamma}_i \right\}$$



Zhou-Athey-Wager (2022) | 1

Extension of AW2021 to multiple treatments

1) Estimate doubly robust policy score (with cross-fitting)

$$\hat{\Gamma}(y_i, d_i, x_i) = \begin{bmatrix} \hat{\Gamma}_0(y_i, d_i, x_i) \\ \vdots \\ \hat{\Gamma}_{M-1}(y_i, d_i, x_i) \end{bmatrix} = \begin{bmatrix} \hat{\mu}(0, x_i) + \dfrac{\underline{1}(d_i = 0)\left[y_i - \hat{\mu}(d_i, x_i)\right]}{\hat{p}(d_i, x_i)} \\ \vdots \\ \hat{\mu}(M-1, x_i) + \dfrac{\underline{1}(d_i = M-1)\left[y_i - \hat{\mu}(d_i, x_i)\right]}{\hat{p}(d_i, x_i)} \end{bmatrix}$$

2) Find best allocation by minimizing regret given the vector of scores

$$\hat{\pi} = \arg\max_{\pi \in \Pi} \left\{ \frac{1}{N} \sum_{i=1}^{N} \sum_{m=0}^{M-1} \underline{1}(\pi(x_i) = m) \hat{\Gamma}_m(y_i, d_i, x_i) \right\}$$



Zhou-Athey-Wager (2022) | Step 1

Conditions

- Sufficient condition: 1st stage estimators of nuisance parameters converge faster than $N^{1/4}$
  - Many ML procedures can achieve this rate under suitable conditions
- Observation *i* must not be used in 1st stage → cross-fitting
  - May be implemented via k-fold cross-validation

This is double machine learning used for policy analysis



Zhou-Athey-Wager (2022) | Step 2

$$\hat{\Gamma}(y_i, d_i, x_i) = \frac{d_i\left[y_i - \hat{\mu}(d_i, x_i)\right]}{\hat{p}(d_i, x_i)} + \begin{bmatrix} \hat{\mu}(0, x_i) \\ \vdots \\ \hat{\mu}(M-1, x_i) \end{bmatrix}$$

Use policy score to find best allocation in restricted class

- Not a convex problem ➔ computationally very demanding, depending on the policy class considered
  - NP-hard
- The simple binary weighted-classification reduction does not carry over directly to the multi-action case
- They provide 2 algorithms for decision trees
  - Exact solution via mixed integer programming
    - Very computationally intensive
  - Tree-search algorithm
    - Exact exhaustive search feasible for shallow trees / moderate problems
    - Approximation possible by restricting candidate splits/search
    - Computationally (**sort-of**) efficient, but does not allow for constraints
    - Cockx, Lechner, Boolens (2023) modify their algorithm to allow for capacity constraints



Which user choices shape an optimal policy tree?

These choices define the score, the feasible policy class, & practical restrictions faced by the algorithm

User-provided parameter	What it does
Decision variables	Which covariates the tree may split on. They determine the information used to differentiate treatment recommendations.
Policy score	The welfare contribution of assigning action *m* to observation *i*. It is the criterion used to compare and value candidate trees.
Tree depth / number of leaves	Controls policy complexity. Deeper trees allow more tailored rules, but increase variance and instability.
Minimum leaf size	Prevents very small terminal groups. This stabilises recommendations and avoids idiosyncratic rules.
Admissible actions	Defines which treatment options may appear in terminal leaves. Some actions can be excluded for legal or institutional reasons.
Constraints	Budgets, capacities, fairness, or eligibility restrictions. They shrink the feasible set of trees.
Welfare criterion / costs	Defines what "optimal" means. Outcomes, costs, and penalties can change which action is best in each leaf.

The estimated tree is (mainly) a joint product of the score, the allowed policy class, & the constraints chosen by the user



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4 | Beyond finding the *best* algorithms

5 | Questions & empirical part



The *best* policy is not just the one with the highest estimated value

A statistically attractive policy may still be problematic in practice.

Value
Does the rule improve expected welfare?

Fairness
Who gains, who loses, and are protected groups treated acceptably?

Implementability
Can caseworkers understand, explain, and apply the rule?

Credibility
Is the policy robust, lawful, and likely to generalise?

Decision algorithms are (should be) evaluated in a broader sense than estimation algorithms



Fairness is also an ethical issue

Fairness questions

Do individuals with similar circumstances receive similar recommendations?

Do protected groups systematically receive less beneficial actions?

Are trade-offs between efficiency & fairness made explicit?

Should some variables be excluded from the decision rule?

Why ethical questions arise

A policy tree does not only predict; it allocates real opportunities.

Recommendations may affect income, training access, or social support.

Even an accurate algorithm may be ethically unacceptable if it reinforces disadvantage or lacks justification.

Hence fairness is not only a technical add-on. It is part of the ethical evaluation of decision rules.



Explainability, legality, & accountability

Explainability

Decision makers should be able to explain why a recommendation is made.

Simple policy trees are attractive because they generate transparent if–then rules.

Legality

Some variables or assignments may be legally restricted.

Public-sector rules must respect eligibility criteria, equal-treatment principles, & due process.

Accountability

Who is responsible if a recommendation is harmful or controversial?

Human oversight and appeals procedures may be necessary.

In many policy settings, a rule with a slightly lower estimated value, but higher transparency, may be preferable



Robustness, uncertainty, & external validity

Main concerns

The optimal tree may be too closely tuned to one sample.

Small data changes may produce different recommendations.

A rule learned in one place or period may not work equally well elsewhere.

Estimated gains are uncertain & should not be treated as exact truths.

Possible responses

Use sample splitting, cross-validation, or repeated splits.

Check whether recommendations are stable across specifications.

Report uncertainty & inspect how sensitive the policy is to modelling choices.

Re-evaluate policies when environments or target populations change.



What else matters for decision algorithms?

Choice of objective
Different objectives can imply different "optimal" policies

Decision variables
The rule depends on which characteristics are allowed to enter

Computational feasibility
Some policy classes are too costly to optimise exactly

Human-in-the-loop
Algorithms may support rather than replace caseworkers

Political legitimacy
A policy rule should be acceptable to stakeholders and institutions

Monitoring after deployment
Policies should be tracked and revised if problems appear



Takeaway: from good algorithms to good decisions

A good decision rule should be …

- effective: it should improve expected welfare
- fair and ethically defensible
- transparent enough to explain and implement
- robust to uncertainty and stable across reasonable modelling choices
- compatible with legal, institutional, and political constraints

The final challenge is not only to find the highest-value rule, but to find a rule society is willing & able to use



Lecture takeaway: from effects to decisions

Optimal policy combines causal knowledge, optimisation, & practical judgement

1 | Why policy learning?

Average effects are often too coarse.

Decisions require knowing which action works best for whom.

2 | General framework

Define policy value & regret.

Estimate policy scores & choose the rule with the highest value.

3 | Optimal policy trees

Restrict attention to simple, interpretable decision rules.

Search for the best full tree within the feasible class.

4 | Beyond the algorithm

Good rules should also be fair, robust, explainable, & implementable.

Legal, ethical, & institutional constraints matter.

Estimate effects → define value → optimise rules → evaluate broader consequences

Thus, the goal is not only to estimate causal effects well, but to turn them into responsible & useful decisions



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Rules for group work (including graded group exercise)

Groups may have up to 4 members

- *All group members must agree to all group members*

Group memberships may change until lecture 3

You are free to use any AI you like

- Graded exam: Contact with humans is prohibited & will be monitored (& will be useless )

Lecturer is available for discussions & help during group work

- limited in graded group work

You should always ***install & test*** the required packages ***before the lecture***

- ***Required (additional) packages are always shown on 1st slide of each lecture***



Empirical group work | 2

The goal is finding an optimal allocation of ALMP to specific unemployed

Use the same data as in lecture 4

All estimations are based on the *mcf* package

- The web documentation contains an example programme that provides all steps needed for effect estimation, optimal decision making, & of-sample-evaluation (good starting point)



Empirical group work | 3

Consider 3 treatments: No treatment, training programme, employment programme

- Aggregate the 2 training & employment programme groups accordingly

Decision maker wants an explainable algorithm that maximises earnings in the 4th quarter of 19X9

- Consider level-2 and level-3 optimal policy trees

Consider 2 sets of not more than 5 decision variables

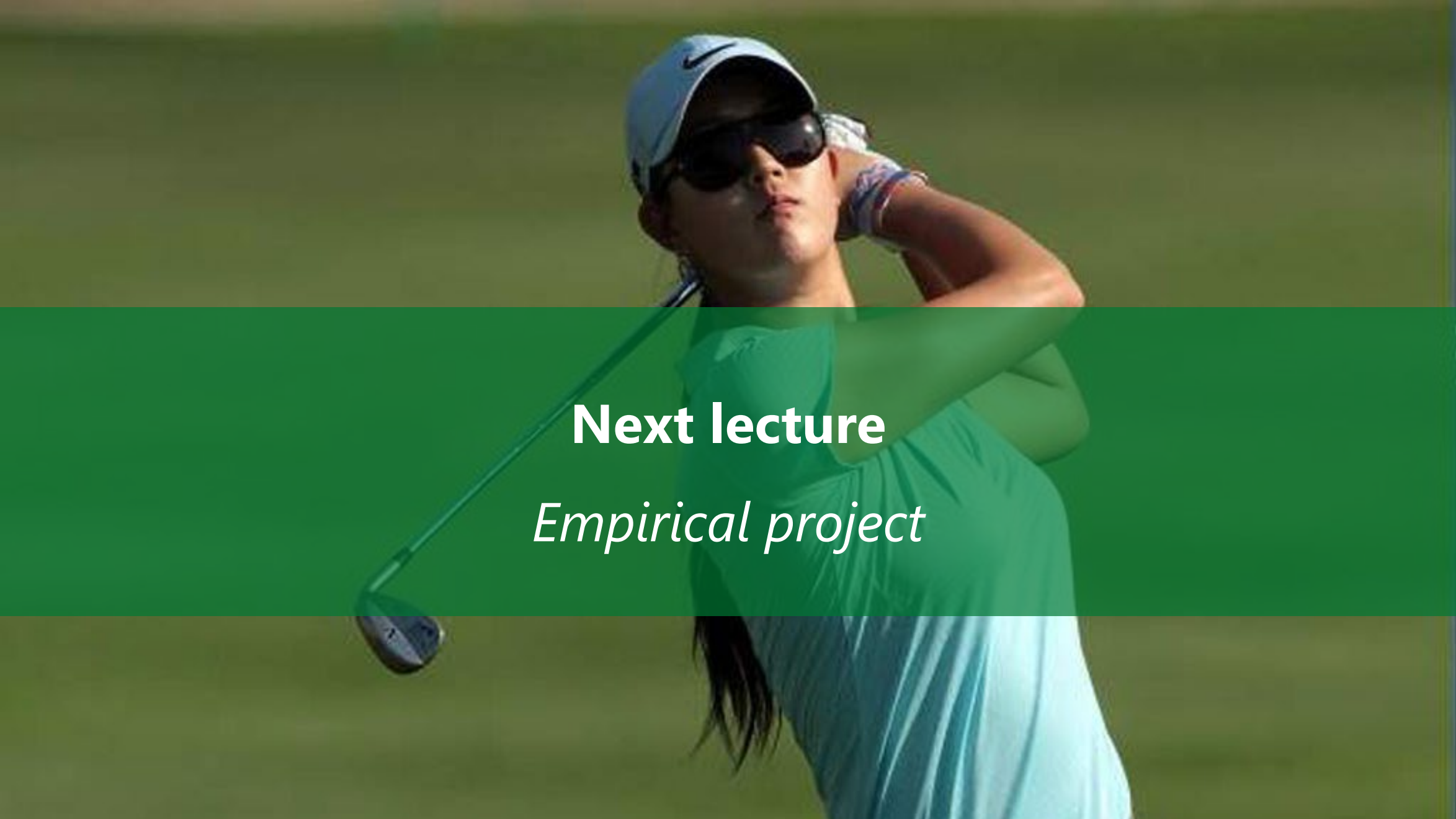
- Select variables according what you consider important & ethically acceptable
- Select variables ignoring the *ethically acceptable* condition



Empirical group work | 4

Characterise the optimal allocations for the 4 different setting

- Programme shares
- Welfare (in-sample vs. out-of-sample)
- Is the choice of decision variables important?
- Is the tree-depth important



Next lecture

Empirical project



Background slides

(for interested students only, not for exam)



More details



Background

Optimal policy & individualised treatment rules | 1

Further remarks

- Major challenges are
  - Estimating expected potential outcomes (or the elements of an alternative policy score) precisely
  - Finding best policy within potentially huge numbers of possible allocations of treatments
    - Could be particularly demanding (statistically & computationally) when there are …
      - » many covariates
      - » many treatments
      - » 'difficult' constraints
      - » dynamics, i.e. sequences of treatments



Optimal policy & minimum regret | 1

$$\pi^* = \arg\max_{\pi \in \Pi} Q(\pi) = \arg\max_{\pi \in \Pi} EY(\pi(X)), \qquad \text{optimal policy}$$

$$R(\pi) = Q(\pi^*) - Q(\pi), \qquad \text{regret}$$

Remarks

- Maximising *Q(π)* is equivalent to minimising *R(π) (minimal regret)*
- Low regret requires sufficiently accurate comparisons of actions, particularly for observations close to the decision boundary
- Nevertheless, this is closer to classification than to regression problems
- Example: $x = a: \quad \mu(1,a) = \mu(0,a); \qquad x = b: \quad \mu(1,b) \ll \mu(0,b)$
  - Optimal policy: **Estimation error for *x=a* does not matter**, for *x=b* it matters hugely
  - Regression-type: They both matter (by how much, depends on distribution of *X*)



Background

Kitagawa-Tetenov (2018) | 1

Binary treatment, known propensity score

Empirical welfare maximisation method

- Maximise sample analogue of average social welfare (sum of potential outcomes)
- Features
  - Constraints on policy
    - Ethical etc. constraints
    - Budget → units per treatment
  - Simple rules based on few 'scores'

Properties for known propensity score

- √N - convergence, *minimax optimality* under fairly general conditions (& unconfoundedness)



Background

Kitagawa-Tetenov (2018) | 2

Structure of estimator

Let $\pi(x)$ be an allocation rule as function of the set of values of x

- This set has lower dimension than x (restricted policies)

The *Empirical Welfare* of a policy is given by

$$W(\pi) = E\left[\frac{DY}{p(X)}\underline{1}(X \in \pi) + \frac{(1-D)Y}{1-p(X)}\underline{1}(X \notin \pi)\right] = EY^0 + E\left[\left(\frac{DY}{p(X)} - \frac{(1-D)Y}{1-p(X)}\right)\underline{1}(X \in \pi)\right]$$

Optimal policy: $$\pi^* = \arg\max_{\pi \in \Pi} W(\pi) = \arg\max_{\pi \in \Pi} E\left\{\left[\frac{DY}{p(X)} - \frac{(1-D)Y}{1-p(X)}\right]\underline{1}(X \in \pi)\right\}$$



Background

Kitagawa-Tetenov (2018) | 3

Estimation: Solve empirical analogue

$$\hat{\pi} = \arg\max_{\pi \in \Pi} W(\pi) = \arg\max_{\pi \in \Pi} \frac{1}{N} \sum_{i=1}^{N} \left[\frac{d_i y_i}{p(x_i)} \underline{1}(x_i \in \pi) + \frac{(1-d_i) y_i}{1-p(x_i)} \underline{1}(x_i \notin \pi) \right]$$

If the link of *x* to the policy is not substantially restricted, this estimation poses huge computational (as well as statistical) challenges (Mixed Integer Linear Programming)



Background

Athey & Wager (2021) | Step 1

$$\hat{\Gamma}_i = \hat{\mu}(1, x_i) - \hat{\mu}(0, x_i) + \frac{d_i - \hat{p}(x_i)}{\hat{p}(x_i)\left[1 - \hat{p}(x_i)\right]}\left[y_i - \hat{\mu}(d_i, x_i)\right]$$

Conditions

- The 3 consistent 1st stage estimators converge at least at rate $N^{1/4}$
  - Many ML procedures achieve this rate (RF, Lasso, some NN, etc.)
- Observation *i* must not be used in 1st stage → cross-fitting
  - May be implemented via k-fold cross-validation

This is double machine learning used for policy analysis



Background

Athey & Wager (2021) | Step 2

Use policy score to find best allocation in restricted class

- Not a convex problem ➔ may be computationally very demanding, depending on the policy class considered (& type & dim of X)
- However, it has an interpretation as a weighted classification problem with sample weights λ_i & response variable H_i

$$\hat{\pi} = \arg\max_{\pi \in \Pi} \left\{ \frac{1}{N} \sum_{i=1}^{N} \lambda_i H_i \left[2\pi(x_i) - 1 \right] \right\}, \quad \lambda_i = \left| \hat{\Gamma}_i \right|, \quad H_i = sign(\hat{\Gamma}_i)$$

- Thus, AW2021 propose to train a (weighted) classifier by standard ML methods



Background

Practical problems | 1

Overfitting

- Optimal policy too closely targeted to sample at hand → may not generalize
  - Split sample randomly → see whether same recommendations emerge in all sample splits (cross-validation type arguments)

Incorporating 'difficult' constraints

- In particular for trees, it seems difficult to introduce them without forgoing efficiency
  - See Cox, Lechner, Boolens (2022) for a proposal



Background

Estimation of IATE and optimal policy: Some 'after-thoughts'

Decision making & effect estimation are different tasks (classification vs. regression)

- Thus, they have different objective functions & which should give better results for the task they are optimised for
- But, if we know potential outcomes (or estimate them without error), assigning treatments according to potential outcomes is optimal (in this case, no need for an only-x-based policy)
- Biases in the policy scores (or consistent estimation of IATEs or potential outcomes) do not matter if these biases are the same for all (individual) potential outcomes
- Thus, in finite samples it *could* be that …
  - Longer data (with some missing confounders) may be preferable to wider data with all confounders (unclear however when this is true)
  - Using precisely estimated potential outcomes (like from the MCF or similar) could give better results than using the DML based policy scores → to be investigated further

Nice discussion in Fernández-Loría, Carlos, Foster Provost (2021): Causal Decision Making and Causal Effect Estimation Are Not the Same… and Why It Matters, arXiv:2104.04103, *INFORMS Journal of Data Science*